



IBM Developer  
SKILLS NETWORK

# Winning Space Race with Data Science

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# Outline

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# Executive Summary

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- The main goal of this project has been to deepen into the data regarding the launches performed by SpaceX, which are available in their public API. To this end, I have performed API requests for data collection, as well as I have performed web scraping in the Wikipedia page corresponding to the SpaceX launches. Once the data were collected, I performed an exploratory data analysis by means of SQL. Moreover, I have used Seaborn package to visually study the data, and to find possible patterns or relationships between different variables, mainly using scatter plots. Deepening into visual analytics, I further performed an analysis using Folium, showing the main space force bases where the launches were carried out, also visualizing the number of successful launches in each station. Finally, I evaluated the performance of different machine learning models, using GridSearchCV to pick up the optimal parameters for each model. Finally, the confusion matrix showcased the capacity of each model to predict the successful launches, comparing their performance to the real data.
- As a result of this complete data analysis, we could elaborate a data frame collecting the major variables extracted from the SpaceX API as well as from the Wikipedia page. Performing the SQL queries, we could get an accurate idea of the nature of the data and observe important correlations among them. Likewise, the use of seaborn allowed us to elucidate certain relationships between variables, for example the tendency to success after certain numbers of flight when the ship was established in certain orbits. The Folium Map also helped us to establish those correlations, even showing that some launch sites had a nearly 100% success rate. Finally, the prediction models displayed an acceptable test score (around 0.83), yet it was evident that every model presented problems with the detection of false positives, thus requiring further implementation. [3](#)

# Introduction

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- **Project background and context**

Since the last century, the aerospace topic has acquired a rising importance in our society, due to its crucial role in the economy and geopolitics of different global powers. Nevertheless, everyone is aware of the high cost of the aerospace missions, which has been a consistent issue for decades. In this context, SpaceX arises with a revolutionary approach: reduce the cost of their missions by the recovery and reuse of the first stage of their rockets, which is something unprecedented. Due to this, it seems worth to analyze the available data of their launches to study the variables that might take place in that first stage recovery

- **Problems you want to find answers**

In this context, Space X has developed the Falcon 9 rocket, which has a cost of 62 million dollars; it seems an acceptable price, since other companies have a cost upward of 165 million dollars for launch. As previously stated, an important percentage of the savings is because Space X can reuse the first stage of their rockets. Thus, if we can determine, with the available data, if the first stage of the rocket will land successfully, we would be able to determine the cost of a launch.



Section 1

# Methodology

# Methodology

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## Executive Summary

- Data collection methodology:
  - Data was collected using *requests* package with the API URL, and obtaining the .json which was then converted to data frame and further cleaned using *Pandas* package.
- Perform data wrangling
  - Data was collected from the corresponding Wikipedia page using again the *requests* package, then a *BeautifulSoup* object was created, which was exported to a list to finally create the data frame.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - Models were built with the *sklearn* package (train\_test split, preprocessing, etc.), parameters were optimized with GridSearchCV and confusion matrix were designed matplotlib.

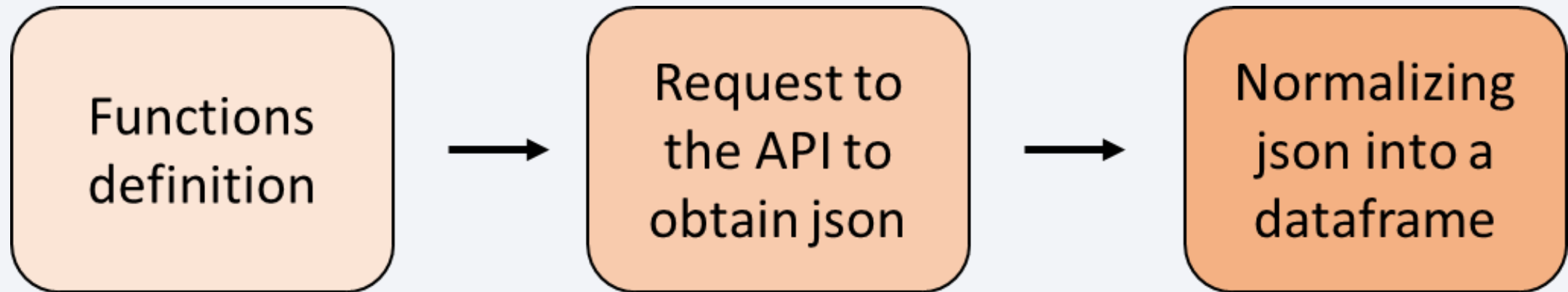
# Data Collection

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- The data collection was performed as follows:
  - Definition of functions for processing the data of the different variables.
  - Request to the URL of the SpaceX API to obtain the json file of interest.
  - Use of pandas for normalization of the json file into a dataframe.
  - Selection of the variables of interest and preprocessing of those variables.
  - Use of the previously defined functions for deeper processing of the data.
  - Filtering to obtain only the data related to the 'Falcon 9' model.
  - Dealing with missing values in the variable 'PayloadMass', replacing them for the mean value.

# Data Collection – SpaceX API

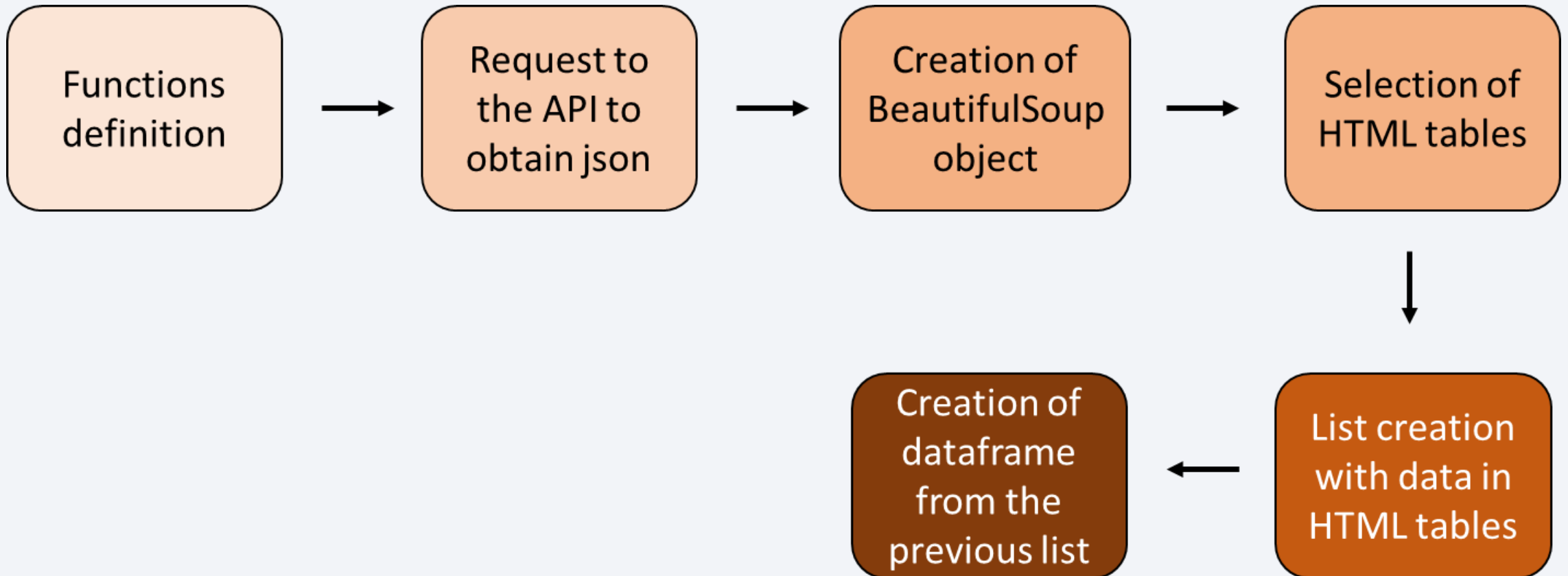
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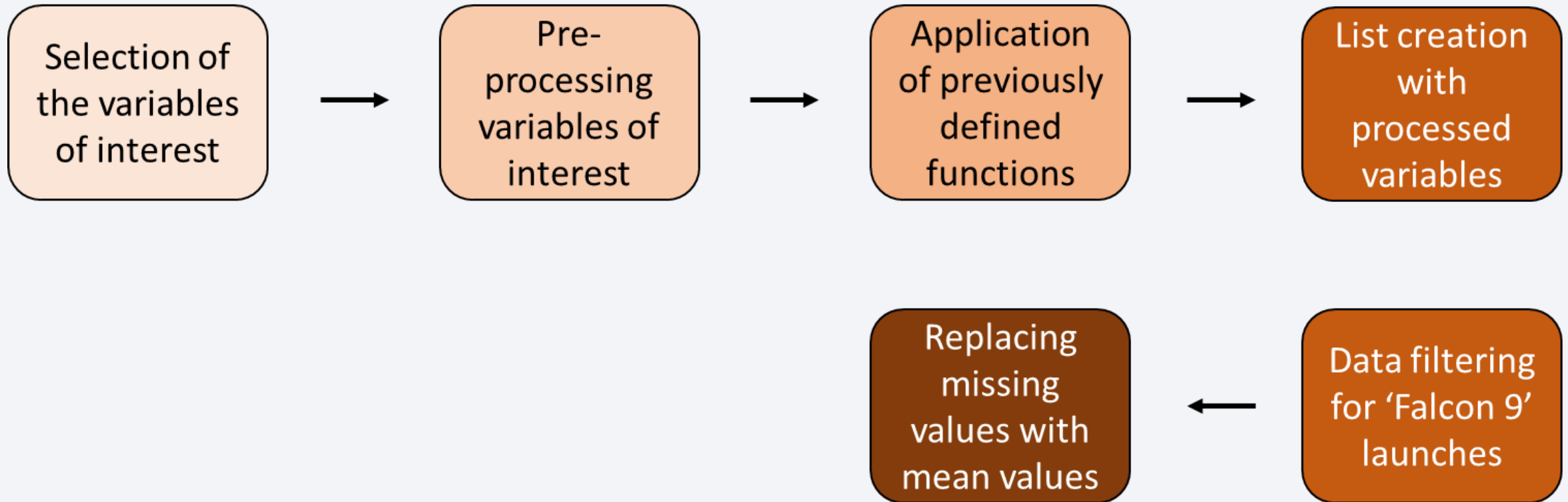
# Data Collection - Scraping

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# Data Wrangling

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# EDA with Data Visualization

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- Summarize what charts were plotted and why you used those charts
- **Scatter plots:** We use this plot to observe the relationship between two different variables, regardless of their nature (categorical or numerical). This way, we can observe, for example, correlation between the launch site and how are the subsequent launches distributed in those places. Besides, we can categorize the data points, assigning a color to them to elucidate, for example, if the launch was successful or not.
- **Bar charts:** We use this cart to quantify categorical values, aiming to compare their value of a given variable among them. For example, we can calculate and represent the success rate of the launches depending on their orbit, and represent that quantification by means of a bar chart.

# EDA with SQL

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Using bullet point format, summarize the SQL queries you performed

- Select unique values of a variable
- Display results that begin with a given text
- Display the total or average value of a variable with certain filters
- Display the first result after applying certain filters
- List the results given after applying several filters
- Using a subquery with a certain aggregate function
- Transform a complete data to month and year variables and apply filters to the year variable
- Performing a count and ranking a given variable using the group by and order by functions

# Build an Interactive Map with Folium

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Summarize what map objects such as markers, circles, lines, etc. you created and added to a folium map. Explain why you added those objects

- **Circle:** Creation of a circle in a specific point with given coordinates. This circle will show a pop-up with a given text whenever it's clicked on.
- **Marker:** Creation of a marker, also for a given pair of coordinates, which can be placed in different points related to a previously created circle. It will show the input text that we included as argument.
- **Cluster:** When we want to show different markers that are placed around the same coordinates, and we don't want them to be overlapped, we can create a cluster to arrange them.
- **Line:** We also created straight lines between two given points, and with the addition of a 'distance function', we can calculate the distance between two points in the map (knowing their coordinates), and we can add them as a marker.



# Build a Dashboard with Plotly Dash

---

Summarize what plots/graphs and interactions you have added to a dashboard. Explain why you added those plots and interactions

- **Pie chart:** The dashboard begins with a pie chart where you can select between the different launch sites where Falcon 9 has been launched. When you do it, you can observe a pie chart with the percentage of successful and failed launches in that launch site. If you select 'All Sites', you can observe the percentage of successful launches in every site with regard to the total number of successful launches.
- **Range slider:** Interactive slider where you can select a range of payload mass values.
- **Scatter plot:** Considering the range of payload mass and launch site chosen, you can observe the distribution of the successful and failed launches with those conditions. Moreover, the data points are colored regarding their Booster Version.

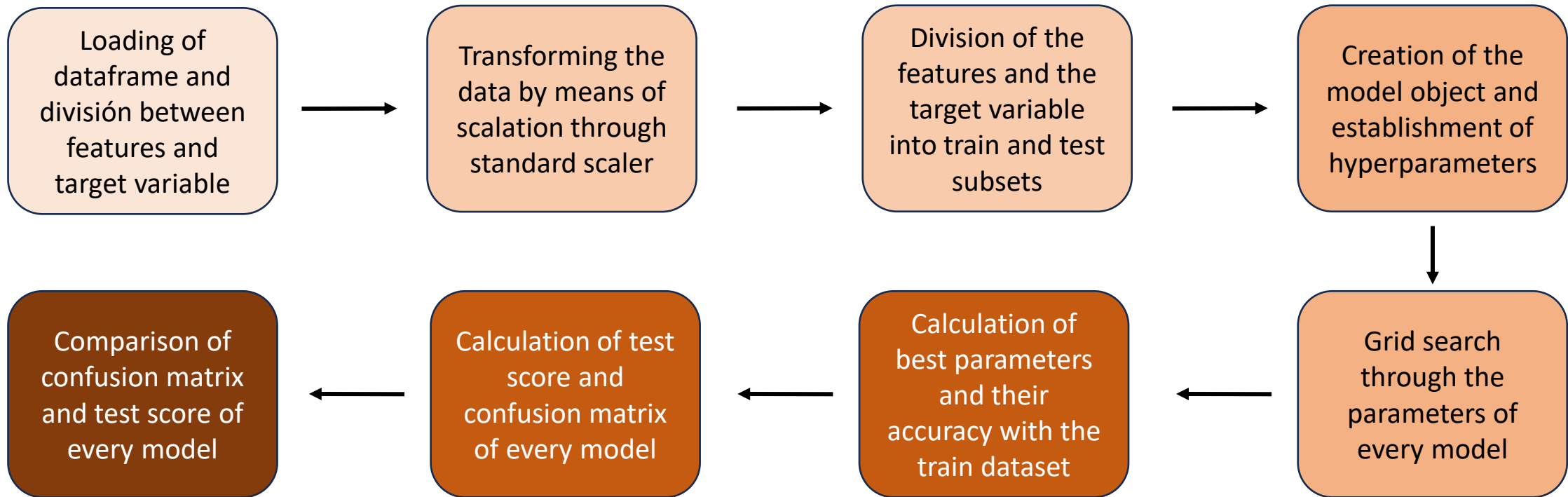
# Predictive Analysis (Classification)

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- Summarize how you built, evaluated, improved, and found the best performing classification model. You need present your model development process using key phrases and flowchart

# Predictive Analysis (Classification)

- Summarize how you built, evaluated, improved, and found the best performing classification model. You need present your model development process using key phrases and flowchart

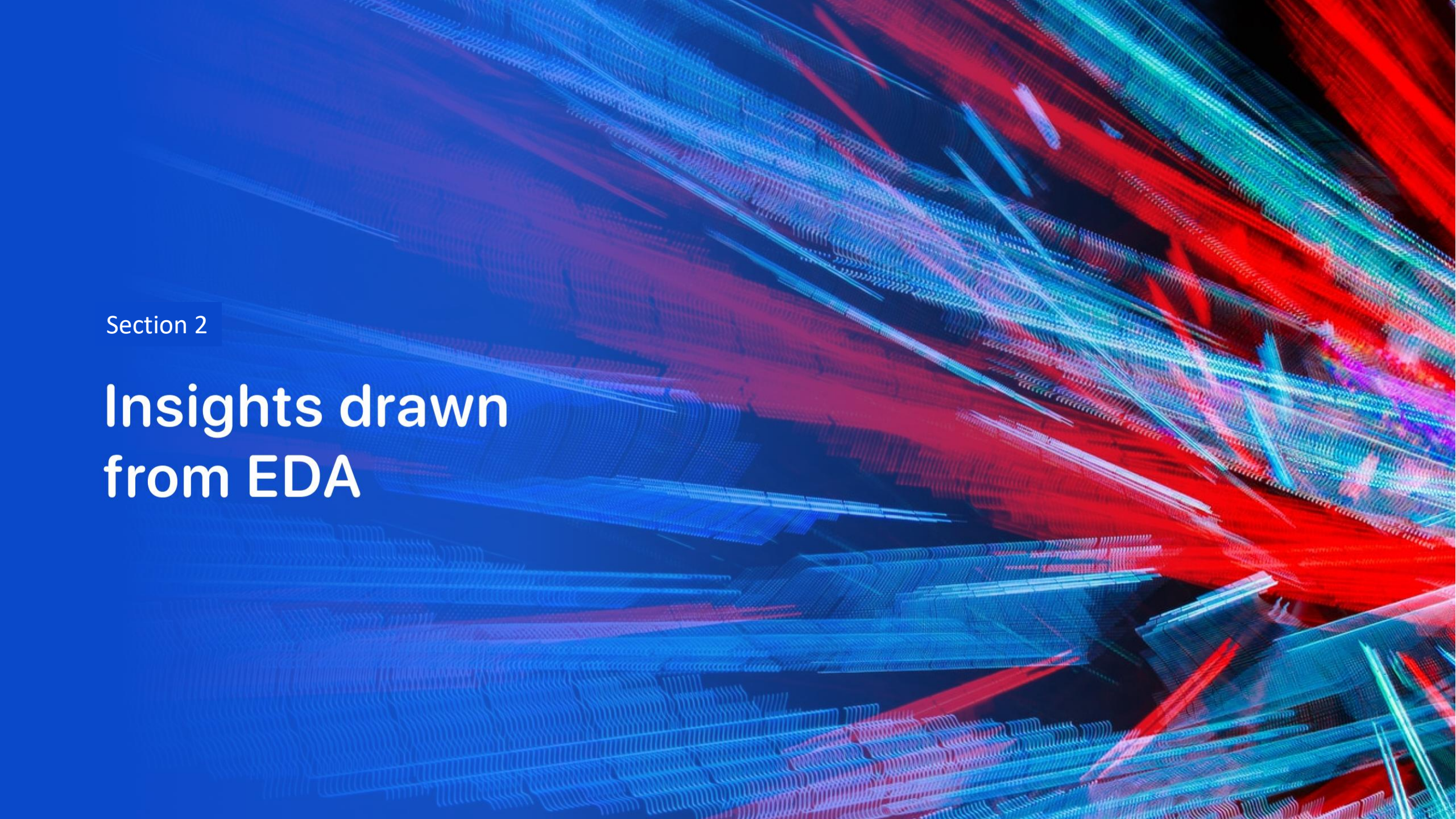


# Results

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- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results



The background of the slide is an abstract composition. It features a dark blue gradient on the left side, which transitions into a complex pattern of diagonal streaks and lines in shades of blue, red, and cyan on the right. These streaks have a textured, almost woven appearance, suggesting a digital or data-driven theme. The overall effect is dynamic and modern.

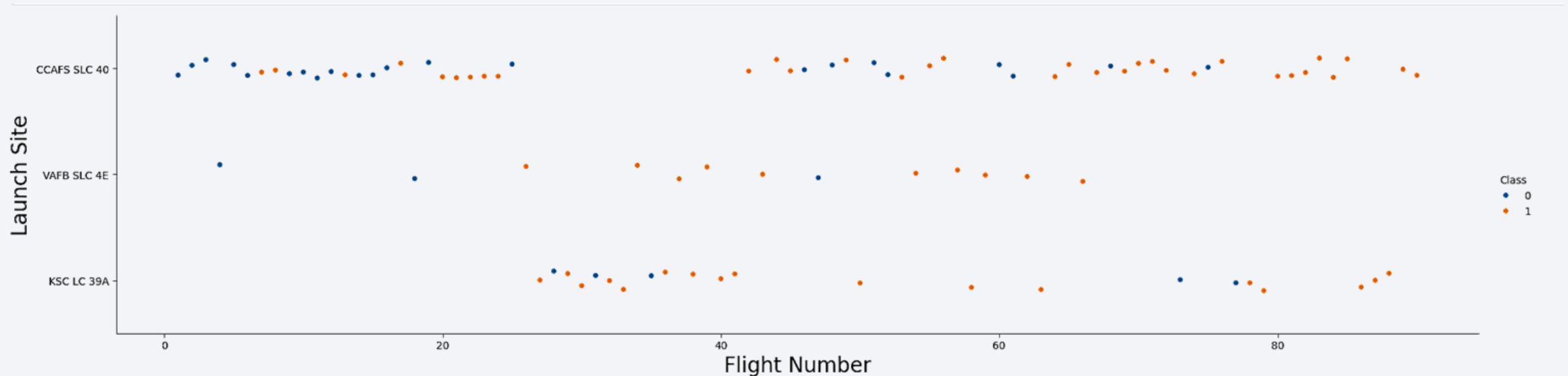
Section 2

# Insights drawn from EDA



# Flight Number vs. Launch Site

- Show a scatter plot of Flight Number vs. Launch Site

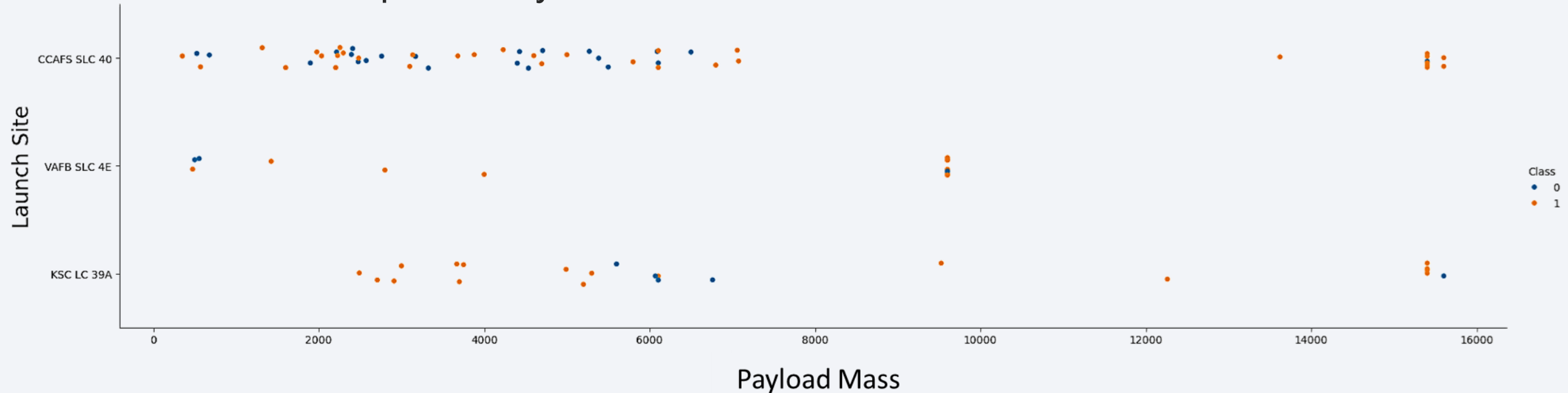


It can be observed, for the launch site CCAFS SLC 40, that as per the Flight Number is higher, the tendency points to a higher efficiency and percentage of successful launches. However, in the rest of launch sites, we don't observe any remarkable trend or tendency.



# Payload vs. Launch Site

- Show a scatter plot of Payload vs. Launch Site

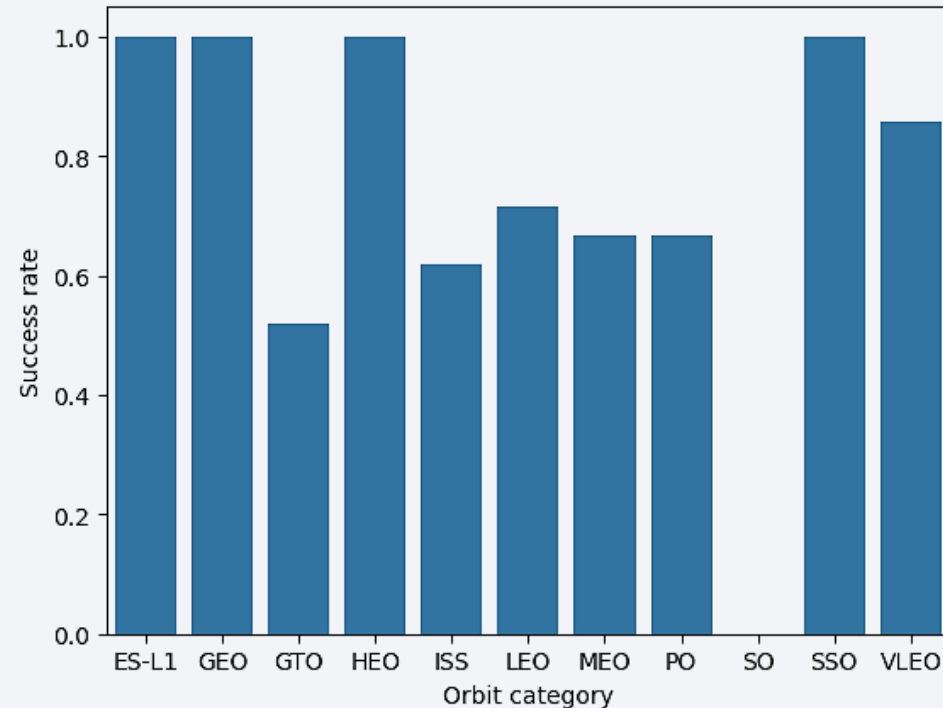


As it can be observed in the launch site CCAFS SLC 40, with higher payload mass it seems that there is a trend pointing to a higher ratio of successful launches. However, this trend cannot be observed in the rest of the Launch sites.

# Success Rate vs. Orbit Type

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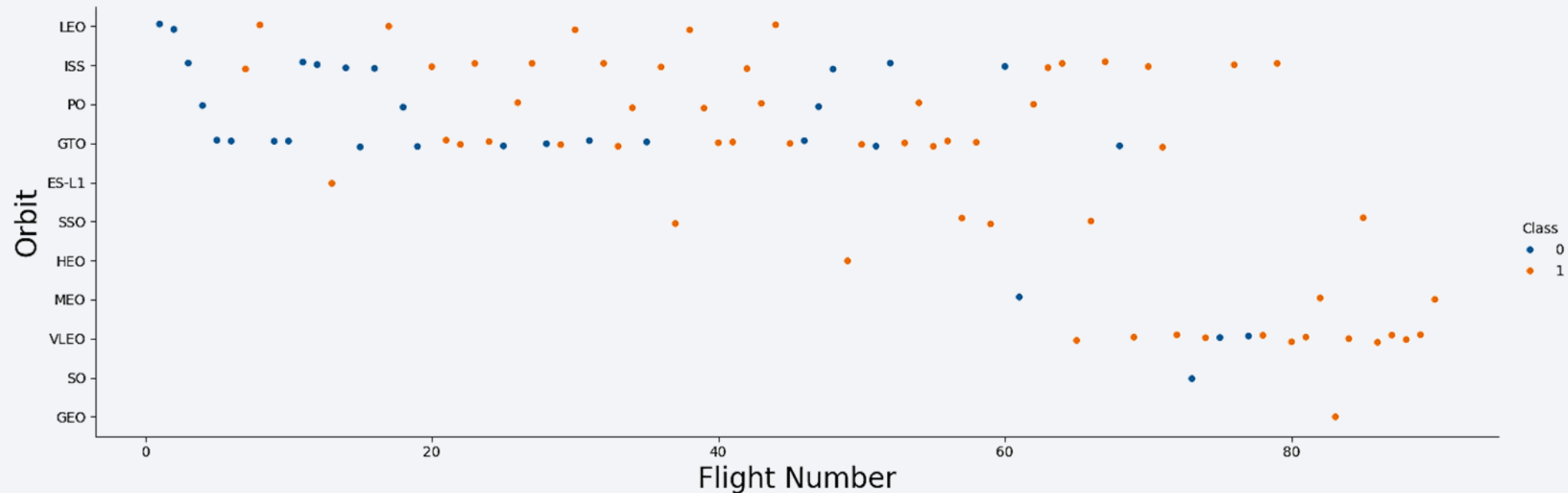
- Show a bar chart for the success rate of each orbit type



- We can observe that some orbit types have a higher success rate than other, such as ES-L1, GEO, HEO or SSO. Other orbits, such as GTO or SO have the lowest success rate.

# Flight Number vs. Orbit Type

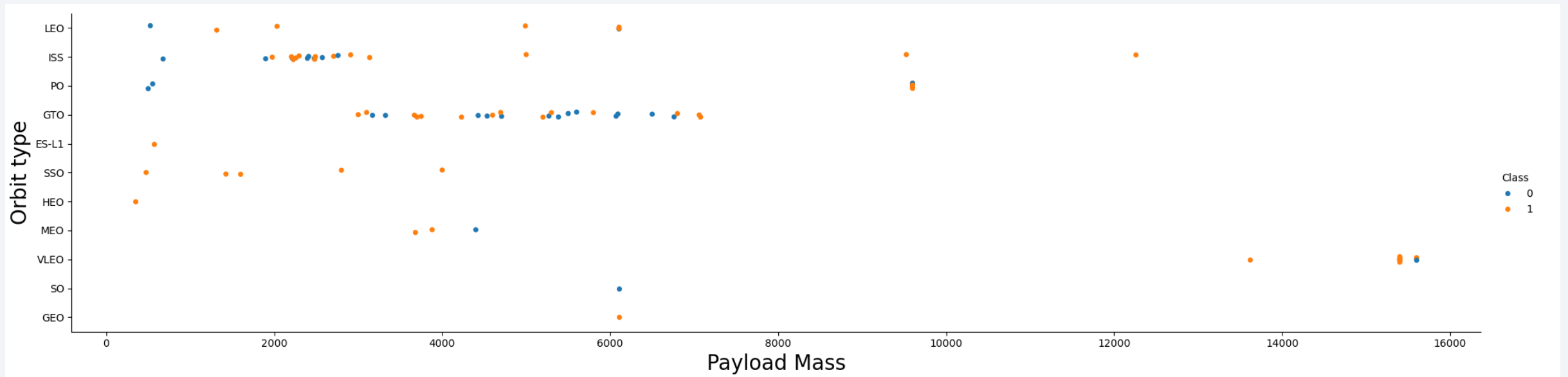
- Show a scatter point of Flight number vs. Orbit type



- This graph displays an interesting event: some of the orbits that presented a success rate of 100% only have a few launches in total, thus the data in those cases is not representative to elucidate a real success rate in those orbits. The higher the number of launches to a certain orbit, the more variability between succesful and failed launches.

# Payload vs. Orbit Type

- Show a scatter point of payload vs. orbit type

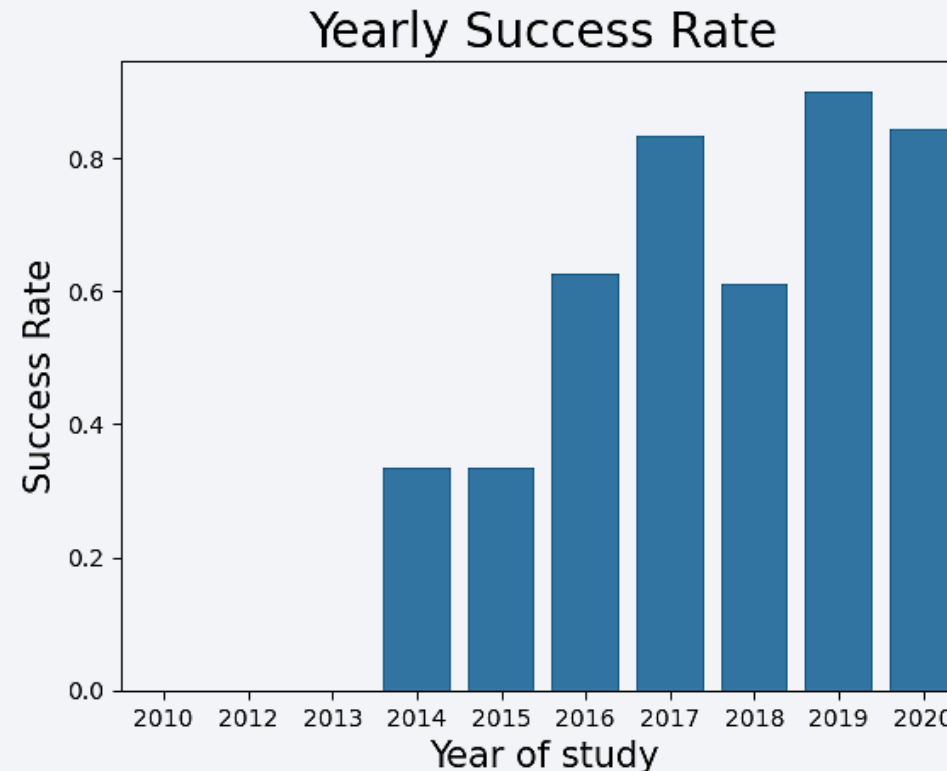


- In this plot we can observe a lot of variability between successful and failed launches for every orbit, regardless of the payload mass value (except for the orbits that don't have enough data points to be considered representative).

# Launch Success Yearly Trend

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- Show a line chart of yearly average success rate



- The success rate established an increasing tren from 2013 to 2020.



# All Launch Site Names

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- Find the names of the unique launch sites

Display the names of the unique launch sites in the space mission

```
%sql SELECT distinct(Launch_Site) FROM SPACEXTABLE;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

| Launch_Site |
|-------------|
|-------------|

|             |
|-------------|
| CCAFS LC-40 |
|-------------|

|             |
|-------------|
| VAFB SLC-4E |
|-------------|

|            |
|------------|
| KSC LC-39A |
|------------|

|              |
|--------------|
| CCAFS SLC-40 |
|--------------|

- With this query, we select the unique Launch Site names from the SpaceX Table

# Launch Site Names Begin with 'CCA'

- Find 5 records where launch sites begin with 'CCA'

```
%sql SELECT * FROM SPACESTABLE WHERE Launch_site like 'CCA%' LIMIT 5;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

| Date       | Time (UTC) | Booster_Version | Launch_Site | Payload                                                       | PAYLOAD_MASS_KG_ | Orbit     | Customer        | Mission_Outcome | Landing_Outcome     |
|------------|------------|-----------------|-------------|---------------------------------------------------------------|------------------|-----------|-----------------|-----------------|---------------------|
| 2010-06-04 | 18:45:00   | F9 v1.0 B0003   | CCAFLC-40   | Dragon Spacecraft Qualification Unit                          | 0                | LEO       | SpaceX          | Success         | Failure (parachute) |
| 2010-12-08 | 15:43:00   | F9 v1.0 B0004   | CCAFLC-40   | Dragon demo flight C1, two CubeSats, barrel of Brouere cheese | 0                | LEO (ISS) | NASA (COTS) NRO | Success         | Failure (parachute) |
| 2012-05-22 | 7:44:00    | F9 v1.0 B0005   | CCAFLC-40   | Dragon demo flight C2                                         | 525              | LEO (ISS) | NASA (COTS)     | Success         | No attempt          |
| 2012-10-08 | 0:35:00    | F9 v1.0 B0006   | CCAFLC-40   | SpaceX CRS-1                                                  | 500              | LEO (ISS) | NASA (CRS)      | Success         | No attempt          |
| 2013-03-01 | 15:10:00   | F9 v1.0 B0007   | CCAFLC-40   | SpaceX CRS-2                                                  | 677              | LEO (ISS) | NASA (CRS)      | Success         | No attempt          |

- With the command Where + 'like' and Limit 5, we select five results that start with the word in quotes (because the % sign is written after the word in quotes)

# Total Payload Mass

---

- Calculate the total payload carried by boosters from NASA

```
%sql SELECT SUM(PAYLOAD_MASS_KG_) FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
SUM(PAYLOAD_MASS_KG_)
```

```
45596
```

- We select the total value (sum()) of payload mass variable, filtering with the where command for the name of the Customer (NASA (CRS))

# Average Payload Mass by F9 v1.1

---

- Calculate the average payload mass carried by booster version F9 v1.1

## Task 4

Display average payload mass carried by booster version F9 v1.1

```
%sql SELECT AVG(PAYLOAD_MASS_KG_) FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

| <u>AVG(PAYLOAD_MASS_KG_)</u> |
|------------------------------|
| 2928.4                       |

- Similarly to the previous task, we calculate the average value of payload mass with the command `avg()`. After that, and using the command `WHERE`, we filter the data to consider only the Booster Version F9 v1.1.

# First Successful Ground Landing Date

- Find the dates of the first successful landing outcome on ground pad

## Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

*Hint: Use min function*

```
%sql SELECT * FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)' ORDER BY Date LIMIT 1;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

| Date       | Time (UTC) | Booster_Version | Launch_Site | Payload                                 | PAYLOAD_MASS_KG_ | Orbit | Customer | Mission_Outcome | Landing_Outcome      |
|------------|------------|-----------------|-------------|-----------------------------------------|------------------|-------|----------|-----------------|----------------------|
| 2015-12-22 | 1:29:00    | F9 FT B1019     | CCAFS LC-40 | OG2 Mission 2 11 Orbcomm-OG2 satellites | 2034             | LEO   | Orbcomm  | Success         | Success (ground pad) |

- We select only the Date variable and filter for the condition (Success (ground pad)). After that, we sort our data by Date using ORDER BY, and we only display the first result

## Successful Drone Ship Landing with Payload between 4000 and 6000

- List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

```
%sql SELECT * FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS_KG_ between 4000 AND 6000;
```

```
* sqlite:///my_data1.db  
Done.
```

| Date       | Time (UTC) | Booster_Version | Launch_Site | Payload               | PAYLOAD_MASS_KG_ | Orbit | Customer               | Mission_Outcome | Landing_Outcome      |
|------------|------------|-----------------|-------------|-----------------------|------------------|-------|------------------------|-----------------|----------------------|
| 2016-05-06 | 5:21:00    | F9 FT B1022     | CCAFS LC-40 | JCSAT-14              | 4696             | GTO   | SKY Perfect JSAT Group | Success         | Success (drone ship) |
| 2016-08-14 | 5:26:00    | F9 FT B1026     | CCAFS LC-40 | JCSAT-16              | 4600             | GTO   | SKY Perfect JSAT Group | Success         | Success (drone ship) |
| 2017-03-30 | 22:27:00   | F9 FT B1021.2   | KSC LC-39A  | SES-10                | 5300             | GTO   | SES                    | Success         | Success (drone ship) |
| 2017-10-11 | 22:53:00   | F9 FT B1031.2   | KSC LC-39A  | SES-11 / EchoStar 105 | 5200             | GTO   | SES EchoStar           | Success         | Success (drone ship) |

- We select the Booster Version variable and we apply our filter (Success (drone ship), and the second filter, which is a payload value between a minimum and a maximum value.



# Total Number of Successful and Failure Mission Outcomes

---

- Calculate the total number of successful and failure mission outcomes

## Task 7

List the total number of successful and failure mission outcomes

```
%sql SELECT COUNT(Mission_Outcome) FROM SPACEXTABLE WHERE Mission_Outcome LIKE 'Fai%';
```

```
* sqlite:///my_data1.db  
Done.
```

| COUNT(Mission_Outcome) |
|------------------------|
| 1                      |

- We select the variable Mission\_Outcome from the table and we perform a counting to display the total values that satisfy the condition, which is to have a Mission\_Outcome class that starts with the word 'Fai'.

# Boosters Carried Maximum Payload

---

- List the names of the booster which have carried the maximum payload mass

## Task 8

List all the booster\_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

```
%sql SELECT DISTINCT(BOOSTER_VERSION) FROM SPACEXTABLE WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);
```

```
* sqlite:///my_data1.db
```

```
Done.
```

| Booster_Version |
|-----------------|
|-----------------|

|               |
|---------------|
| F9 B5 B1048.4 |
|---------------|

|               |
|---------------|
| F9 B5 B1049.4 |
|---------------|

|               |
|---------------|
| F9 B5 B1051.3 |
|---------------|

|               |
|---------------|
| F9 B5 B1056.4 |
|---------------|

|               |
|---------------|
| F9 B5 B1048.5 |
|---------------|

|               |
|---------------|
| F9 B5 B1051.4 |
|---------------|

|               |
|---------------|
| F9 B5 B1049.5 |
|---------------|

|               |
|---------------|
| F9 B5 B1060.2 |
|---------------|

|               |
|---------------|
| F9 B5 B1058.3 |
|---------------|

|               |
|---------------|
| F9 B5 B1051.6 |
|---------------|

|               |
|---------------|
| F9 B5 B1060.3 |
|---------------|

|               |
|---------------|
| F9 B5 B1049.7 |
|---------------|

- We select the unique values of the variable BOOSTER\_VERSION that satisfy the condition of the subquery. This condition is to present the maximum value in the payload mass variable

# 2015 Launch Records

---

- List the failed landing\_outcomes in drone ship, their booster versions, and launch site names for in year 2015

## Task 9

List the records which will display the month names, failure landing\_outcomes in drone ship ,booster versions, launch\_site for the months in year 2015.

**Note: SQLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.**

```
%sql SELECT substr(Date, 6,2), Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTABLE WHERE Landing_Outcome = 'Failure (drone ship)' AND substr(Date,0,5)='2015';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

|    | substr(Date, 6,2)    | Landing_Outcome | Booster_Version | Launch_Site |
|----|----------------------|-----------------|-----------------|-------------|
| 01 | Failure (drone ship) | F9 v1.1 B1012   | CCAFS LC-40     |             |
| 04 | Failure (drone ship) | F9 v1.1 B1015   | CCAFS LC-40     |             |

- We divide the variable Date into year and monthm then we select the variables to display: month, Landing\_Outcome, Booster\_Version and Launch\_Site. After that we establish both our conditions with the command Where to filter the data.

# Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

## Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

```
%sql SELECT Landing_Outcome, COUNT(Landing_Outcome) FROM SPACEXTABLE WHERE Date between '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY COUNT(Landing_Outcome) DESC;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

| Landing_Outcome        | COUNT(Landing_Outcome) |
|------------------------|------------------------|
| No attempt             | 10                     |
| Success (drone ship)   | 5                      |
| Failure (drone ship)   | 5                      |
| Success (ground pad)   | 3                      |
| Controlled (ocean)     | 3                      |
| Uncontrolled (ocean)   | 2                      |
| Failure (parachute)    | 2                      |
| Precluded (drone ship) | 1                      |

- We select the variable of interest (Landing\_Outcome), and we establish a new column for the count values of this variable. Then we establish the condition for filtering with the where command, we group the data to perform the count on the same variable, and finally we order the data by the new established variable (Count(Landing\_Outcome))

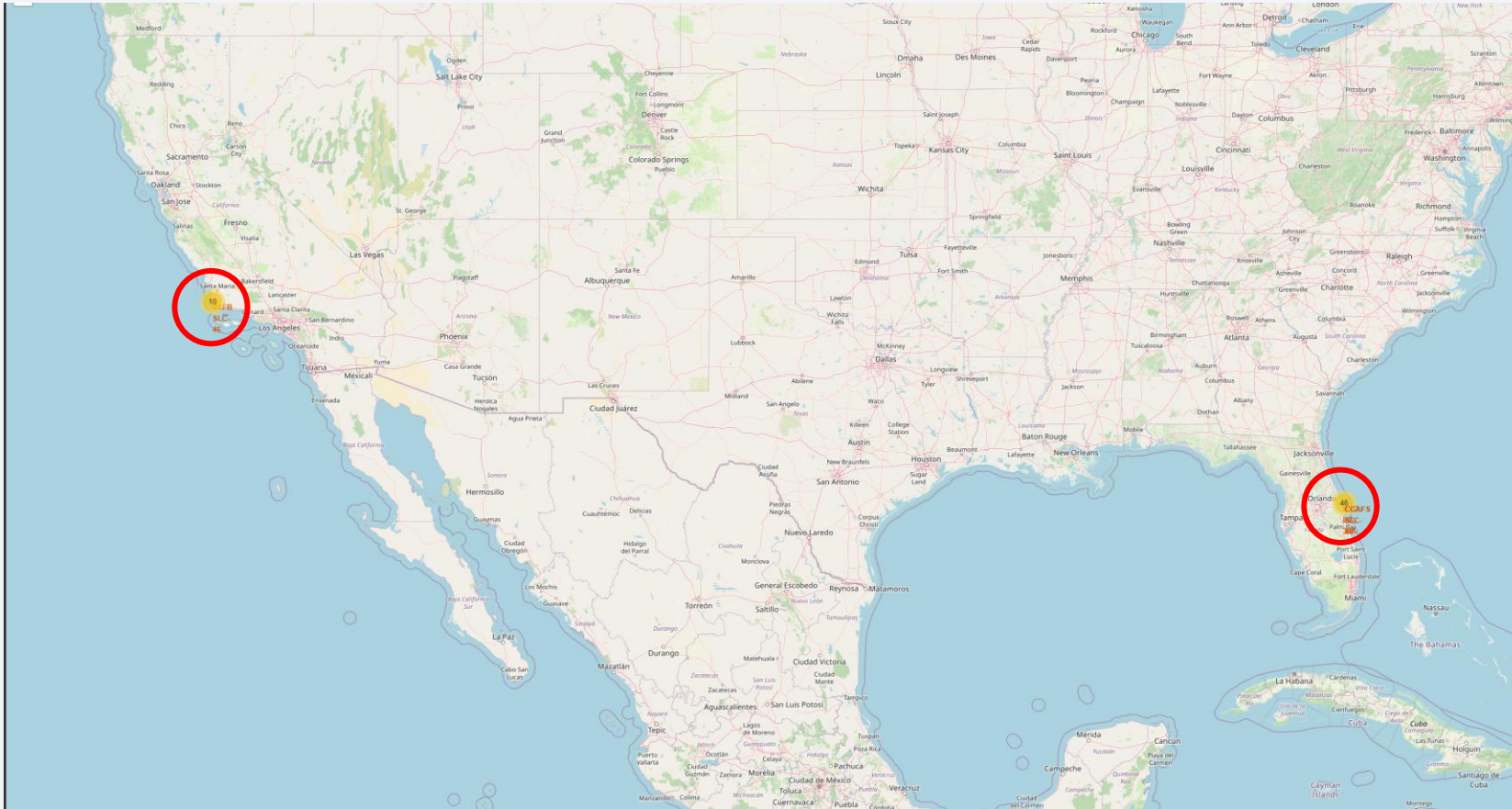
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

# Launch Sites Proximities Analysis

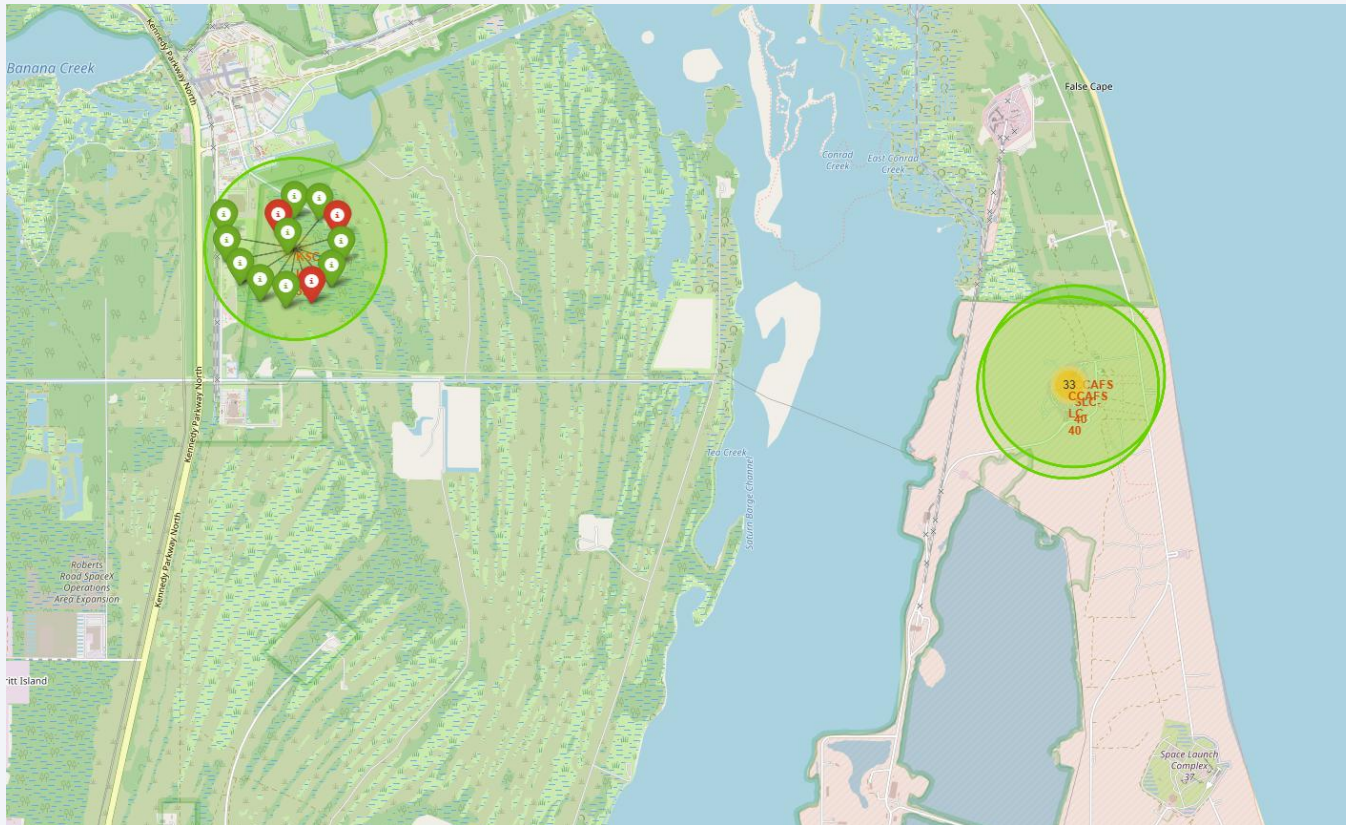


# Launch site locations in Folium Map



- As it can be observed in the map, two red circles have been drawn in the points corresponding to the space force bases indicated with the circles and markers in Folium

# Color-labeled Launch outcomes in a Launch site

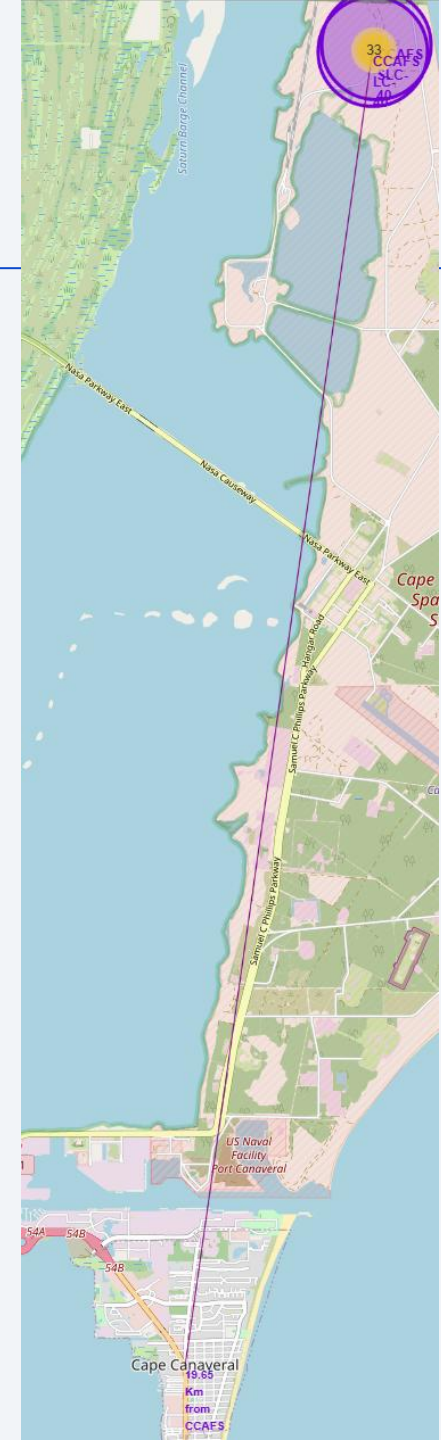


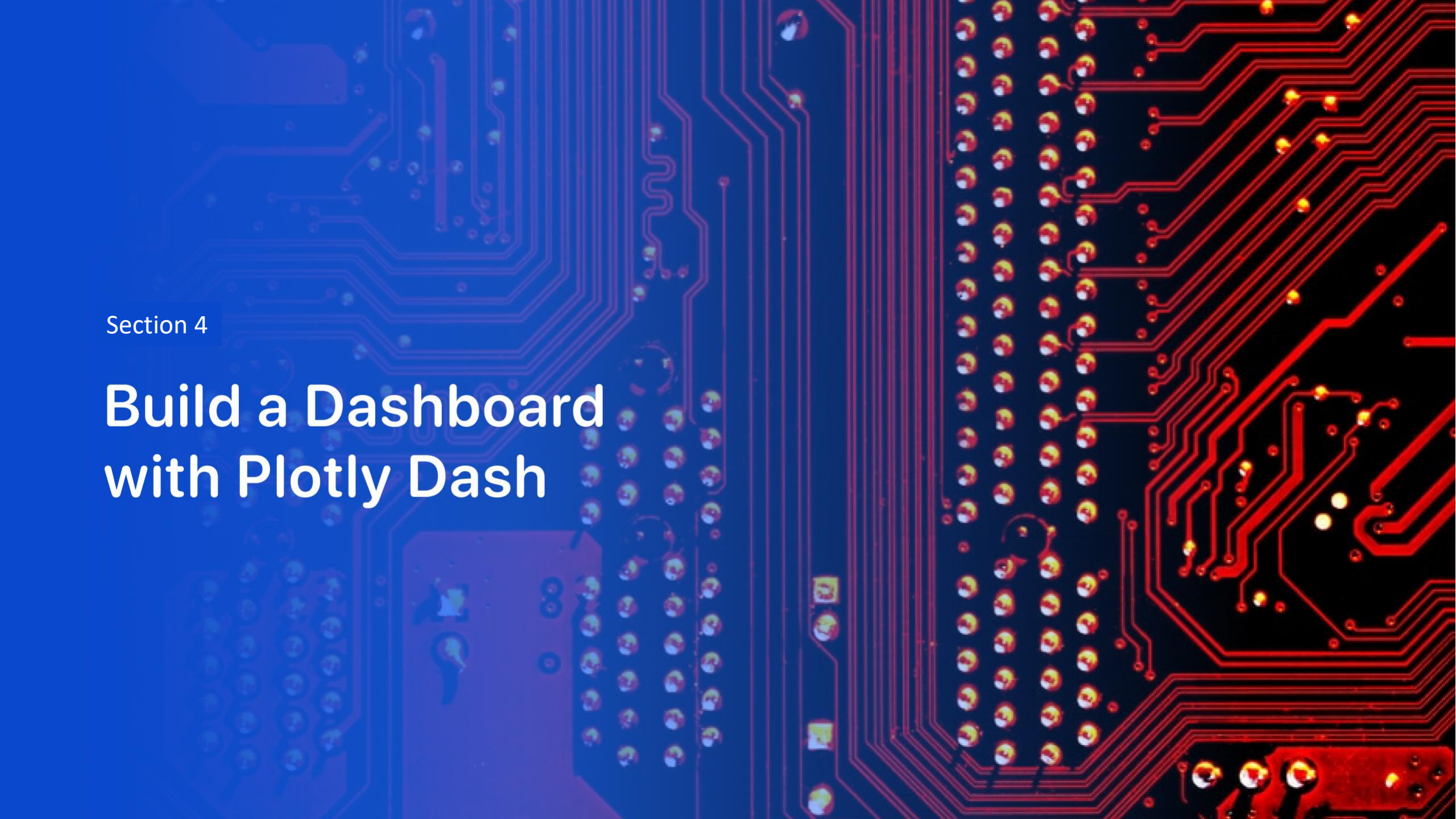
- As it can be observed in the map, the cluster of markers corresponding to the different launch outcomes has been created and represented in the map.



# Distance from two points

- As it can be observed in the map, the distance between two points has been drawn. In particular, a line from the space force base CCAFS SLC-40 and Cape Canaveral has been drawn



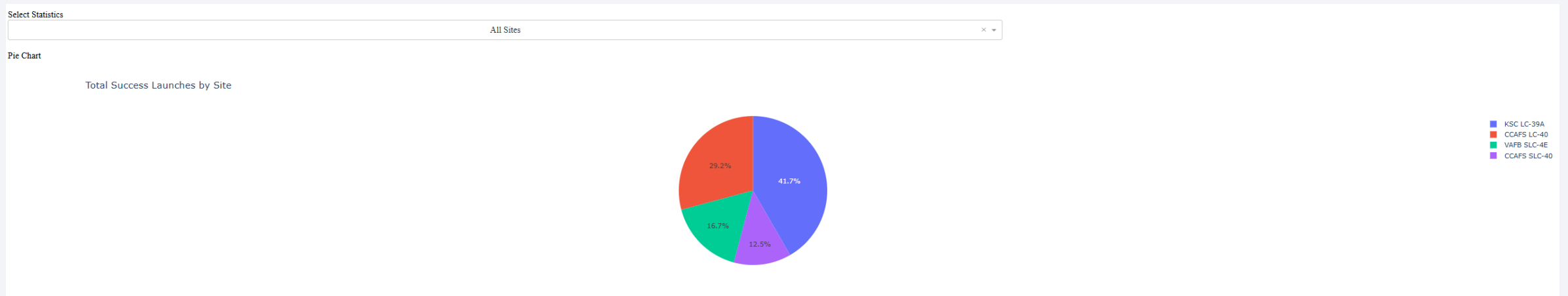


Section 4

# Build a Dashboard with Plotly Dash

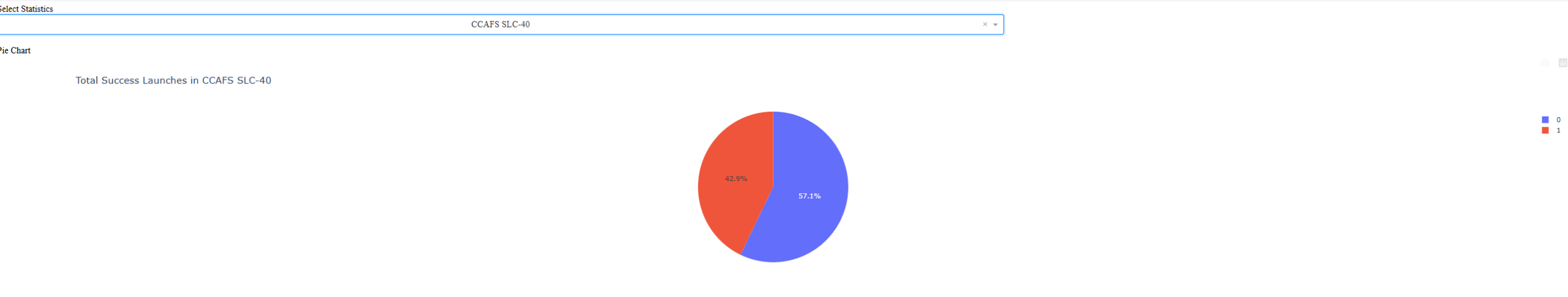
# Launch success count for all sites

- Explain the important elements and findings on the screenshot



- The pie chart shows the percentage of success launches for every launch site, in their corresponding percentage related to the total number of successful launches

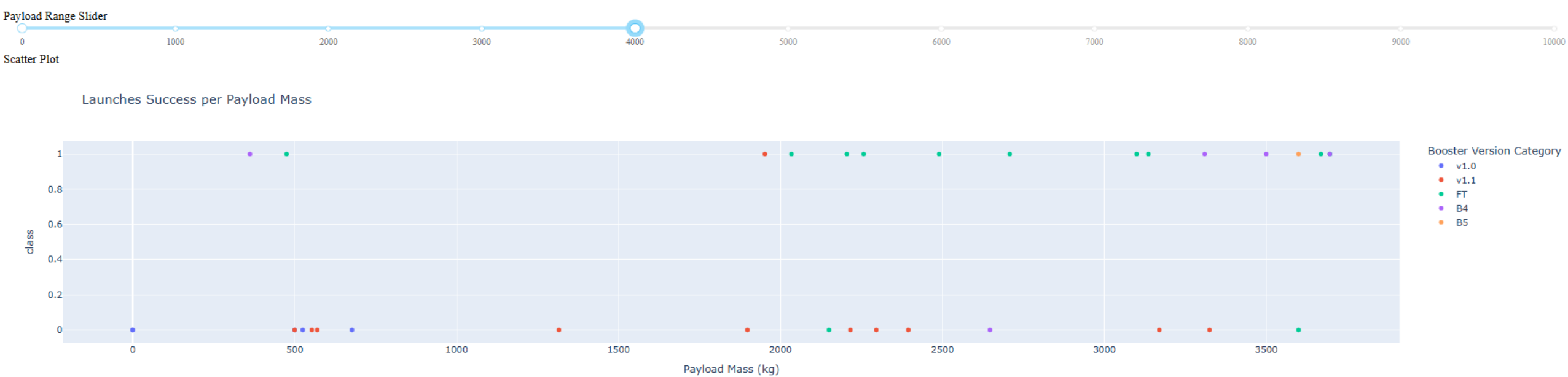
# Launch site with the highest success ratio



- The launch site with the highest success ratio is CCAFS SLC-40, with a 42.9% of successful launches

# <Dashboard Screenshot 3>

- Explain the important elements and findings on the screenshot, such as which payload range or booster version have the largest success rate, etc.



# <Dashboard Screenshot 3>

- Explain the important elements and findings on the screenshot, such as which payload range or booster version have the largest success rate, etc.





## <Dashboard Screenshot 3>

Explain the important elements and findings on the screenshot, such as which payload range or booster version have the largest success rate, etc.

- The most important elements in the figure are the range slider, which allows to study different ranges of payload mass; and also the scatter plot generated, which shows the successful launches in that range, establishing different colors for the different booster versions
- Regarding the payload range, it seems that there is no correlation between different values of payload and the launch success possibilities. Regarding the booster version, although there is no version that exhibits an outstanding performance, it seems that the FT version has a higher ratio of successful launches, whereas v1.1 and B5 present a very low ratio of success.

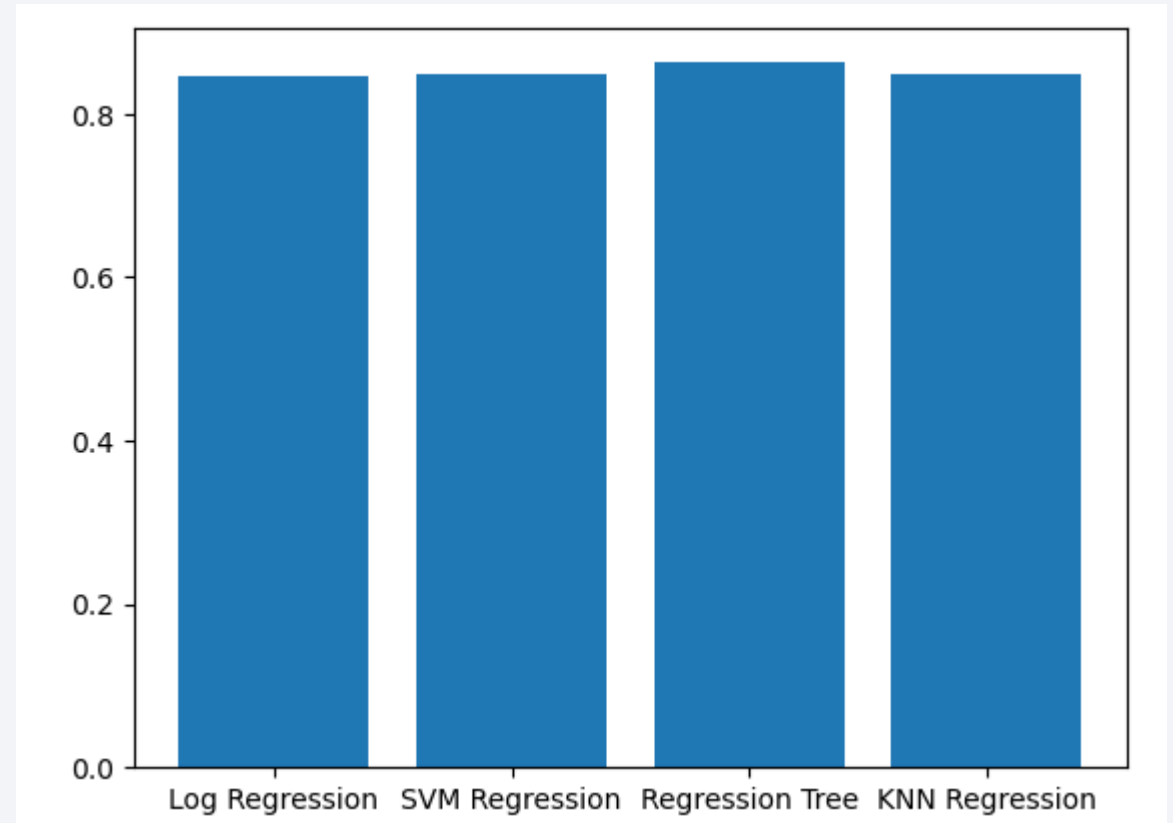
Section 5

# Predictive Analysis (Classification)

# Classification Accuracy

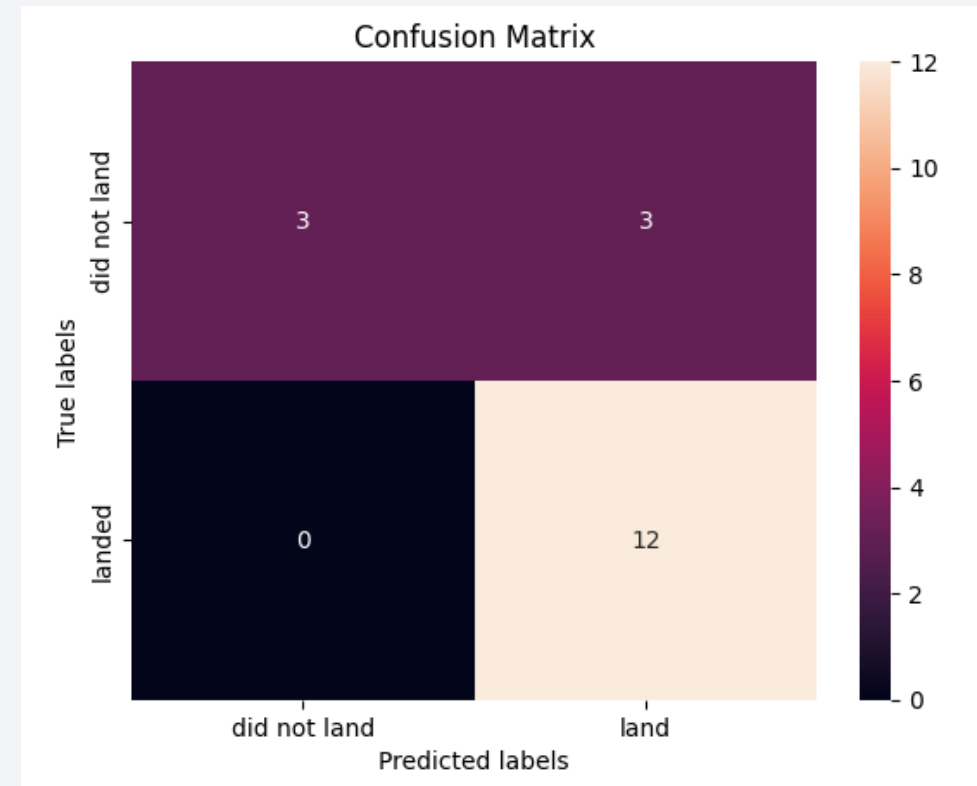
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- As it can be observed, all the models have a quite similar value of accuracy, only showing non-significant differences among them.



# Confusion Matrix

- Show the confusion matrix of the best performing model with an explanation



- Every model performed exactly the same in the confusion matrix, presenting problems with the detection of false positives

# Conclusions

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- For the launch site CCAFS SLC 40, that as per the Flight Number is higher, the tendency points to a higher efficiency and percentage of successful launches
- There is a lot of variability between successful and failed launches for every orbit, regardless of the payload mass value
- The different launch sites present different ratios of successful launches, compared to their total launches for every station.
- The launch site with the highest success ratio is CCAFS SLC-40, with a 42.9% of successful launches
- Although there is no version that exhibits an outstanding performance, it seems that the FT version has a higher ratio of successful launches, whereas v1.1 and B5 present a very low ratio of success.
- All the machine learning models studied in this project have a quite similar value of accuracy, only showing non-significant differences among them.

# Appendix

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- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project



Thank you!

